EMBODIED MODEL DISAGREEMENT PROTOCOLS: TURNING WORLD-MODEL UNCERTAINTY INTO PHYSICAL PROBES

Robotics 60 Paper Batch, Paper 60
Recovery build
`embodied_model_disagreement_protocols`

ABSTRACT

Robot world models increasingly expose disagreement through ensembles, Bayesian heads, uncertainty penalties, or rollout variance. This paper argues that embodied agents should not treat this disagreement primarily as a scalar confidence score. For robots, the operational question is: what small physical probe would distinguish the candidate models without making the scene unsafe? We propose an embodied model disagreement protocol that compiles a disagreement set into executable probe actions, predicted observation signatures, safety gates, and stopping rules. A 1200-row literature sweep shows that the local field is rich in uncertainty, planning, and world-model work, with planning-control as the largest discovered tag cluster. In a deterministic diagnostic over 1080 trials, executable probes resolve 0.795 of disagreements and reduce unsafe false accepts to 0.106, compared with 0.347 for a scalar uncertainty threshold. The contribution is deliberately modest: a protocol-level mechanism and an inspectable witness, not a claim of hardware dominance.

1 MOTIVATION

Disagreement among robot world models is often useful, but it is under-specified. A high ensemble variance can tell a planner that something is uncertain, yet it does not say whether the robot should tap the object, lift a corner, change viewpoint, pause for tactile evidence, or avoid the action entirely. That missing compilation step matters because embodied uncertainty is not just epistemic; it is tied to contact, support, occlusion, compliance, and safety.

The central thesis is that model disagreement should be turned into a physical question. Instead of asking whether the ensemble is confident enough, the robot asks which low-cost observation would distinguish the candidate dynamics while respecting a safety gate. This converts uncertainty from a passive score into an executable protocol.

2 RELATED WORK

The recovery sweep collected 1200 rows in `docs/related_work_matrix.csv`, then produced 300-paper, 225-paper, and 100-paper review subsets. The largest tag cluster was planning-control with 498 rows. The sweep supports a narrow gap: the field already has ensemble dynamics, learned world models, active learning, exploration, and safe planning, but these lines often stop before specifying the physical observation that would resolve a particular embodied disagreement.

World models and model-based control provide predictive candidates for planning (Ha and Schmidhuber, 2018; Nagabandi et al., 2018). Ensembles and uncertainty estimates help identify where those candidates diverge (Lakshminarayanan et al., 2017; Chua et al., 2018). Active learning and exploration select information-gathering actions (Settles, 2009). Robotics adds the missing constraint: the query is a physical intervention with contact and safety consequences (Levine et al., 2016; Brohan et al., 2023).

Figure 1: Deterministic diagnostic over six embodied ambiguity classes. Executable probes resolve more model disagreements and reduce unsafe false accepts, but pay a deliberate physical-probe cost.

Table 1: Mean diagnostic results. The protocol improves disagreement resolution and reduces unsafe false accepts at the cost of executing physical probes.

Method	Resolved	Unsafe false accepts	Probe cost
Uncertainty threshold	0.472	0.347	0.171
Latent disagreement	0.614	0.267	0.335
Executable probe protocol	0.795	0.106	1.927

3 Protocol

Let $\mathcal{M} = \{m_i\}$ be candidate embodied world models and b be the current belief over them. For a candidate probe action a, each model predicts an observation signature $o_i(a)$: visual change, tactile impulse, support transition, or short-horizon state change. The protocol chooses

$$a^\star = \arg\max_{a \in \mathcal{A}_{safe}} \Big[D(\{o_i(a)\}) - \lambda C(a) - \beta R(a) \Big], \tag{1}$$

where D is predicted model-separating disagreement, C is physical cost, R is risk, and $\mathcal{A}_{safe}$ is filtered by a safety gate. After executing the probe, the robot updates b, discards models whose predicted signatures disagree with the observation, and either continues probing or returns to task planning.

The protocol has four required outputs:

1. an executable probe action;
2. predicted observation signatures for each candidate model;
3. a safety gate and abort condition;
4. a stopping rule for when the disagreement is sufficiently resolved.

These outputs distinguish the protocol from a scalar uncertainty bonus.

4 Diagnostic

We implement a deterministic witness with six ambiguity classes: free-space dynamics, occluded contact, friction cones, support graphs, tool hinges, and object identity. Three evaluation styles are compared: a scalar uncertainty threshold, a latent disagreement score, and the executable probe protocol. The diagnostic produces 1080 trials in `docs/probe_protocol_trials.csv` and summary rows in `docs/probe_protocol_results.csv`.

The result is not surprising, and that is the point. If a disagreement is about friction, contact, support, or identity, passive uncertainty scores often detect ambiguity without resolving it. The executable protocol spends physical cost to ask a discriminating question. Figure 1 and Table 1 make the tradeoff explicit: the protocol is not free, but it reduces the dangerous case where the planner accepts a model that should have been tested.

5 Limitations

This paper should be read as a protocol proposal. The diagnostic is synthetic, the probe library is hand-specified, and no hardware claim is made. The largest unresolved problem is the safety gate: in difficult manipulation settings, deciding whether a probe is safe may be as hard as the original planning problem. The protocol is still useful because it forces papers to expose that requirement rather than hiding it behind a scalar uncertainty score.

6 Conclusion

Embodied model disagreement is valuable only when it changes what the robot does. The proposed protocol turns disagreement into safe physical probes with predicted observation signatures and stopping rules. For the batch artifact, the paper supplies a literature matrix, review subsets, diagnostic data, source, figure, and compiled PDF. For future work, the obvious next step is a hardware study where the same disagreement set is tested with visual, tactile, and contact probes.

References

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